MIL-C-62/4B
19 December 1968
SUPERSEDING
62/4A
25 November 1964

MILITARY SPECIFICATION SHEET

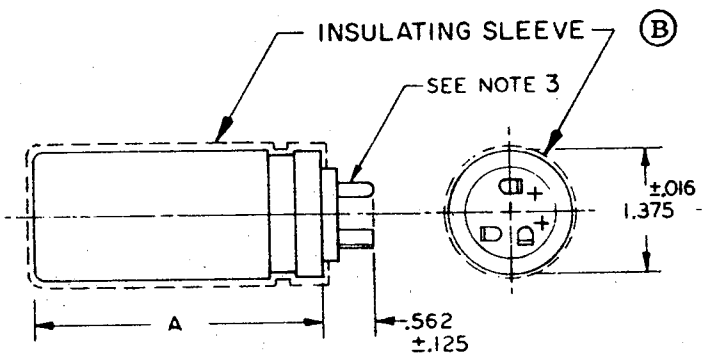
CAPACITORS, FIXED, ELECTROLYTIC

(DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED)

STYLES CE32 AND CE35

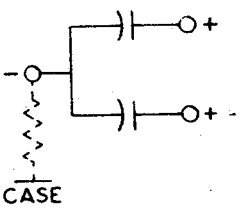
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-62.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.016	.41
.125	3.18
.562	14.27
1.375	34.93

STYLES
CE32 – uninsulated
CE35 – insulated



UNINSULATED STYLE NOT FOR NEW DESIGN.
FOR REPLACEMENT PURPOSES ONLY.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Shape and position of lugs optional.
4. Capacitors are mounted by means of a mounting ring. (Family 2).
5. There is an indeterminate resistance between the metal case and the negative terminal.

Ⓑ denotes changes

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FSC 5910

Ⓑ

TABLE I. Styles CE32 and CE35 capacitors.

Type designation 4/	Capacitance	DC rated voltage per section	DC leakage current per section	Maximum impedance per section at 120 $+\frac{10}{-5}$ Hz at -40° $+\frac{0°}{-3°}$ C	Dissipation factor per section	Case dimension 1/3/ A ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE3-C680J	68-68	150	151	129	12	2.250 (57.15)
CE3-C151J	150-150	150	225	59	12	3.000 (76.20)
CE3-C221J	220-220	150	273	40	12	3.750 (95.25)
CE3-C620K	62-62	200	166	142	12	2.250 (57.15)
CE3-C820K	82-82	200	192	107	12	2.750 (69.85)
CE3-C101K	100-100	200	211	88	12	2.750 (69.85)
CE3-C151K	150-150	200	260	59	12	3.250 (82.55)
CE3-C300M	30-30	250	129	293	12	2.250 (57.15)
CE3-C470M	47-47	250	166	187	12	2.250 (57.15)
CE3-C560M	56-56	250	178	157	12	2.500 (63.50)
CE3-C680M	68-68	250	195	129	12	2.750 (69.85)
CE3-C750M	75-75	250	206	117	12	3.000 (76.20)
CE3-C101M	100-100	250	237	88	12	3.250 (82.55)
CE3-C151M	150-150	250	290	59	12	3.250 (82.55)
CE3-C360N	36-36	300	156	244	12	2.250 (57.15)
CE3-C680N	68-68	300	214	129	12	3.250 (82.55)
CE3-C101N	100-100	300	260	88	12	4.250 2/ (107.95)
CE3-C151N	150-150	300	318	59	12	4.250 2/ (107.95)
CE3-C100P	10-10	350	88	880	12	2.250 (57.15)
CE3-C200P	20-20	350	126	440	12	2.250 (57.15)
CE3-C300P	30-30	350	154	293	12	2.250 (57.15)
CE3-C470P	47-47	350	192	187	12	2.750 (69.85)
CE3-C620P	62-62	350	222	142	12	3.250 (82.55)
CE3-C680P	68-68	350	232	129	12	3.750 (95.25)
CE3-C101P	100-100	350	280	88	12	4.250 2/ (107.95)

See footnotes at end of table.

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(B) TABLE I. Styles CE32 and CE35 capacitors. (Continued)

Type designation 4/	Capacitance	DC rated voltage per section	DC leakage current per section	Maximum impedance per section at 120 $+^{10}_{-5}$ Hz at -40° $+^{0°}_{-3°}$ C	Dissipation factor per section	Case dimension 1/3/ A ±.062 (1.57)
	uf	volts	ua	ohms	percent	inches
CE3-C100Q	10-10	400	94	1480	17	2.250 (57.15)
CE3-C150Q	15-15	400	116	986	17	2.250 (57.15)
CE3-C220Q	22-22	400	141	670	17	2.250 (57.15)
CE3-C300Q	30-30	400	163	493	17	2.750 (69.85)
CE3-C330Q	33-33	400	172	450	17	3.000 (76.20)
CE3-C470Q	47-47	400	206	314	17	3.500 (88.90)
CE3-C150R	15-15	450	123	1465	17	2.250 (57.15)
CE3-C200R	20-20	450	142	1100	17	2.500 (63.50)
CE3-C240R	24-24	450	156	915	17	2.750 (69.85)
CE3-C300R	30-30	450	174	733	17	3.500 (88.90)
CE3-C430R	43-43	450	208	512	17	3.750 (95.25)
CE3-C470R	47-47	450	217	468	17	4.250 2/ (107.95)

1/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

2/ Tolerance on this dimension shall be +.062 (1.57), -.187 (4.75) inch.

3/ Dimensions shown are for style CE32; for style CE35, add .062 (1.57) ±.062 (1.57) to the nominal dimension.

4/ Complete type designation shall include an additional digit to complete style.

REQUIREMENTS:

Dimensions and configuration:
See figure and table I.

Case type: Tubular.

Capacitance value: See table I.

Capacitance tolerance:
At 200 volts, dc rated, and above:
-10, +50 percent.
At 150 volts, dc rated, and less:
-10, +75 percent.

Operating temperature range:
-40° C to +85° C.

(B) Solderability: Method 208 of MIL-STD-202.

DC leakage: See table I.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: See table I.

Terminal strength: Method 211 of MIL-STD-202, test conditions A, B, and E.

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Shock, medium impact: Method 205, test condition C, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Temperature cycling and immersion:
Temperature cycling: Method 102, test condition D, MIL-STD-202.
Immersion: Method 104, test condition C, MIL-STD-202.
DWV-Insulating sleeves-Method 301 of, MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial measurement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Moisture resistance: Method 106 of MIL-STD-202.
Insulation resistance (insulating sleeve) - Not less than 100 megohms.
DWV-Insulating sleeves-Method 301 of MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial requirement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Stability at low and high temperature:
In accordance with MIL-C-62.
Step 1 (at +25° C):
DCL - See table I.
Capacitance - See table I.
DF - See table I.

Step 2 (at -40° C):
Impedance - See table I.

Step 3 (at +25° C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Step 4 (at +85° C):
DCL - Not more than 10 times initial value, or meet initial 25° C requirement.
Capacitance - ±20 percent of initial value.
DF - See table I.

Step 5 (at +25° C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Surge voltage: In accordance with MIL-C-62.
DCL - See table I.
Capacitance change - 6 percent.
DF - See table I.

Vent: Not applicable.

Life: Method 108, test condition F, MIL-STD-202.
DCL - See table I.
Capacitance change:
100 volts, dc rated and less: -20, +15 percent.
DC rated voltage above 100 volts: ±15 percent.
DF - Meet the initial requirement or not exceed 150 percent of the initial measurement.

Barometric pressure: Method 105 of MIL-STD-202.

Shelf life: In accordance with MIL-C-62.
Build factor: Not more than 4.0.
Shelf factor: Not more than 3.0.
DCL: Not more than 150 percent of initial value.
Capacitance change: 6 percent.
DF: Not more than 175 percent of initial requirement.

Application data:
All capacitors shall withstand an rms ripple current at 120 Hz and 85° C as determined from the formula:

$$I_r = k\sqrt{C}$$

Where:

I_r = rms ripple current in milliamps.

C = nominal capacitance in microfarads per section.

k = 50 for units rated at 150 and 200 volts dc.

= 35 for units rated at 250 and 350 volts dc.

= 30 for units rated at 400 and 450 volts dc.

Ⓑ Supersession data: See table II.

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Ⓑ TABLE II. Supersession data.

Superseded part number	Replacement style CE32 or CE35	Superseded part number	Replacement style CE32 or CE35
001	CE3-C100P	028	CE3-C360N
002	CE3-C300M	029	CE3-C470P
003	CE3-C470M	030	CE3-C620P
004	CE3-C680J	031	CE3-C680N
005	CE3-C820K	032	CE3-C101N
006	CE3-C101K	033	CE3-C151N
007	CE3-C151J	034	CE3-C100P
008	CE3-C221J	035	CE3-C200P
009	CE3-C100P	036	CE3-C300P
010	CE3-C300M	037	CE3-C470P
011	CE3-C470M	038	CE3-C470P
012	CE3-C470M	039	CE3-C620P
013	CE3-C620K	040	CE3-C680P
014	CE3-C820K	041	CE3-C101P
015	CE3-C101K	042	CE3-C100Q
016	CE3-C151K	043	CE3-C150Q
017	CE3-C100P	044	CE3-C220Q
018	CE3-C300M	045	CE3-C300Q
019	CE3-C300M	046	CE3-C330Q
020	CE3-C470M	047	CE3-C470Q
021	CE3-C560M	048	CE3-C150R
022	CE3-C680M	049	CE3-C150R
023	CE3-C750M	050	CE3-C200R
024	CE3-C101M	051	CE3-C240R
025	CE3-C151M	052	CE3-C300R
026	CE3-C100P	053	CE3-C430R
027	CE3-C300P	054	CE3-C470R

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-62/4B Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized) Styles CE32 and CE35

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 56 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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